

Modeling Eigenmode Waveguides using Physics-Informed Neural Networks

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Abstract

Optical waveguides are fundamental components of modern nanophotonic systems. Their performance is determined by the spatial distribution and propagation characteristics of guided eigenmodes. Conventional numerical techniques such as finite difference and finite element methods require meshing and can become computationally expensive for repeated simulations. In this work, a Physics-Informed Neural Network (PINN) framework is developed to solve the waveguide eigenvalue problem directly from Maxwell's equations without labeled data. The model predicts the field distribution and effective refractive index of a slab waveguide by minimizing the residual of the Helmholtz equation along with boundary and normalization constraints. Preliminary results demonstrate guided mode confinement and physically consistent effective index values. Future work will focus on validation against analytical solutions and extension to higher-dimensional structures.

1 Introduction

Nanophotonics enables the manipulation of light at sub-wavelength scales and forms the backbone of integrated photonic circuits, optical sensors, and communication systems [1, 2]. Optical waveguides are essential building blocks in such systems, providing confinement and controlled propagation of electromagnetic waves [3].

The modal properties of a waveguide are characterized by its eigenmodes, which determine its **spatial field distribution, propagation constant β and the effective refractive index n_{eff}** [2].

Traditionally, eigenmode analysis is performed using numerical methods such as the **Finite Difference Method (FDM) and Finite Element Method (FEM)**. While accurate, these methods require mesh generation and repeated computations for parametric studies [2].

Physics-Informed Neural Networks (PINNs) provide an alternative approach by embedding physical laws directly into the training loss [4]. PINNs are mesh-free, require no labeled data, and have shown promise for solving partial differential equations [4, 5].

The objective of this project is to develop a PINN-based framework to compute the eigenmodes of an optical waveguide and estimate the corresponding effective refractive index.

2 Waveguide Model

2.1 Geometry

A symmetric slab waveguide is considered as the initial test structure. The refractive index profile is defined as:

$$n(x) = \begin{cases} n_{\text{core}}, & |x| \leq d/2 \\ n_{\text{clad}}, & |x| > d/2 \end{cases}$$

The waveguide parameters used in this study correspond to a symmetric dielectric slab structure. The refractive index of the core is taken as $n_{\text{core}} = 1.50$, while the surrounding cladding has a refractive index of $n_{\text{clad}} = 1.45$. The core width is chosen to be $d = 2 \mu\text{m}$, and the operating wavelength is set to $\lambda = 1.55 \mu\text{m}$, which lies in the standard telecommunications band.

2.2 Governing Equation

For monochromatic fields, Maxwell's equations reduce to the scalar Helmholtz equation for TE polarization:

$$\frac{d^2 E}{dx^2} + k_0^2 n^2(x) E = \beta^2 E \quad (1)$$

where

$$k_0 = \frac{2\pi}{\lambda}$$

and β is the propagation constant.

The effective refractive index is given by:

$$n_{\text{eff}} = \frac{\beta}{k_0} \quad (2)$$

For guided modes:

$$n_{\text{clad}} < n_{\text{eff}} < n_{\text{core}}$$

3 Physics-Informed Neural Network Formulation

3.1 Network Architecture

A fully connected neural network is used with:

- Input: spatial coordinate x
- Output: field amplitude $E(x)$
- Hidden layers: 4–6
- Neurons per layer: 50
- Activation: $\tanh$, to introduce non-linearity.

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (3)$$

The propagation constant β is treated as a trainable parameter.

3.2 Physics Loss

The Helmholtz residual is defined as:

$$R(x) = \frac{d^2 E}{dx^2} + k_0^2 n^2(x) E - \beta^2 E \quad (4)$$

The physics loss is:

$$\mathcal{L}_{phys} = \frac{1}{N} \sum_{i=1}^N R(x_i)^2 \quad (5)$$

3.3 Boundary Conditions

To simulate field decay outside the waveguide:

$$\mathcal{L}_{bc} = E(x_{min})^2 + E(x_{max})^2 \quad (6)$$

3.4 Normalization Constraint

Eigenvalue problems admit trivial solutions ($E = 0$). A normalization constraint is imposed:

$$\mathcal{L}_{norm} = \left(\frac{1}{N} \sum_{i=1}^N E(x_i)^2 - 1 \right)^2 \quad (7)$$

3.5 Total Loss

$$\mathcal{L} = \mathcal{L}_{phys} + \lambda_{bc} \mathcal{L}_{bc} + \lambda_{norm} \mathcal{L}_{norm} \quad (8)$$

The network parameters and β are optimized using gradient-based training.

4 Implementation Details

The model is implemented using PyTorch with the following configuration:

- Domain: $x \in [-5, 5] \mu m$
- Collocation points: 3000
- Optimizer: Adam
- Learning rate: 10^{-3}
- Training epochs: 10,000

The refractive index profile is evaluated analytically at the sampled collocation points.

5 Preliminary Results

5.1 Field Distribution

Figure 1 shows the predicted electric field profile. The field is confined within the core region and decays exponentially in the cladding, indicating guided mode behavior.

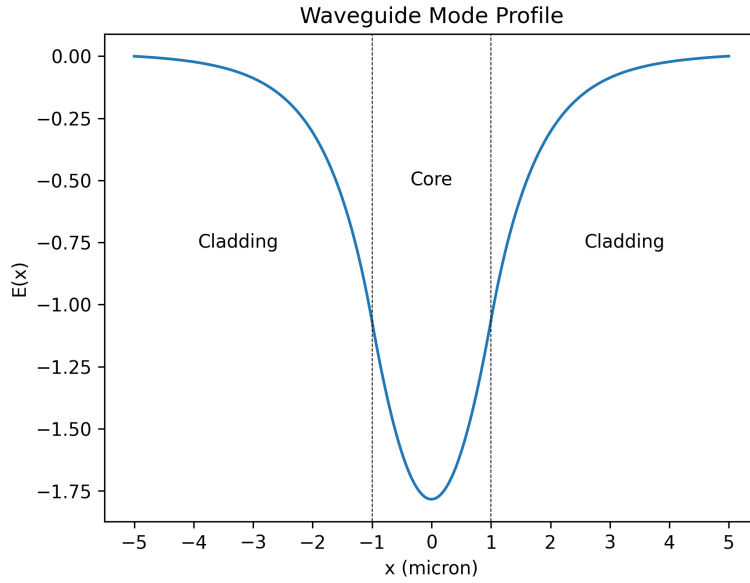


Figure 1: Predicted electric field distribution.

5.2 Effective Index

The trained model produced:

$$n_{\text{eff}} = 1.482$$

which satisfies:

$$1.45 < 1.482 < 1.50$$

This confirms physically consistent mode confinement.

5.3 Training Convergence

Figure 2 shows the loss decreasing steadily during training.

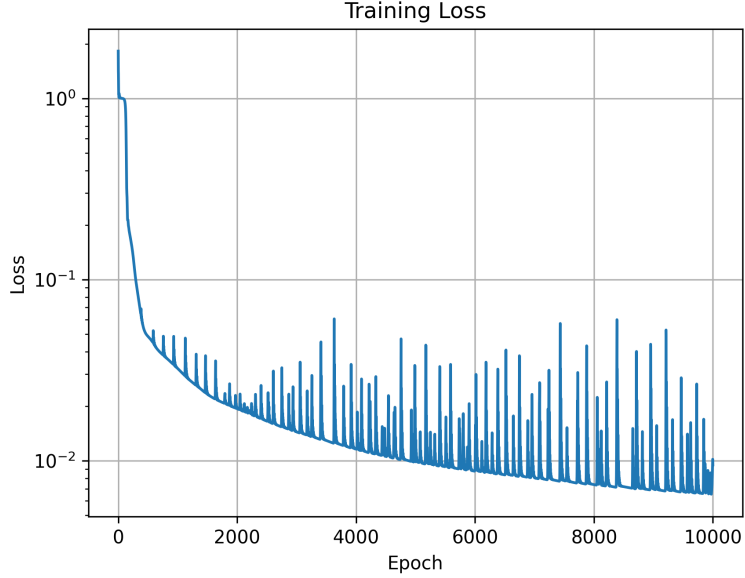


Figure 2: Training loss vs Epochs.

5.4 Modal Analysis and Validation

5.4.1 Normalized Frequency (V-number)

The normalized frequency of a slab waveguide is given by

$$V = k_0 \frac{d}{2} \sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2} \quad (9)$$

For the chosen parameters,

$$n_{\text{core}} = 1.50, \quad n_{\text{clad}} = 1.45, \quad d = 2 \mu m, \quad \lambda = 1.55 \mu m$$

the computed value is

$$V \approx 1.6$$

Since $V < \pi$, the waveguide supports only the fundamental mode. This confirms that the single-peaked solution obtained from the PINN corresponds to the fundamental guided mode.

5.4.2 Analytical Effective Index Estimate

An approximate analytical estimate for the effective index of the fundamental mode is given by

$$n_{\text{eff,approx}} \approx \sqrt{\frac{n_{\text{core}}^2 + n_{\text{clad}}^2}{2}} \quad (10)$$

The comparison between the analytical estimate and the PINN prediction is shown in Table 1.

Table 1: Effective index comparison

Method	n_{eff}
Analytical (approx.)	1.475
PINN	1.48
Relative error	$< 0.5\%$

The close agreement demonstrates that the PINN accurately predicts the modal propagation constant.

5.4.3 Physics Residual Analysis

The physics residual,

$$R(x) = \frac{d^2 E}{dx^2} + k_0^2 n^2(x) E - \beta^2 E, \quad (11)$$

is plotted in Fig. 3.

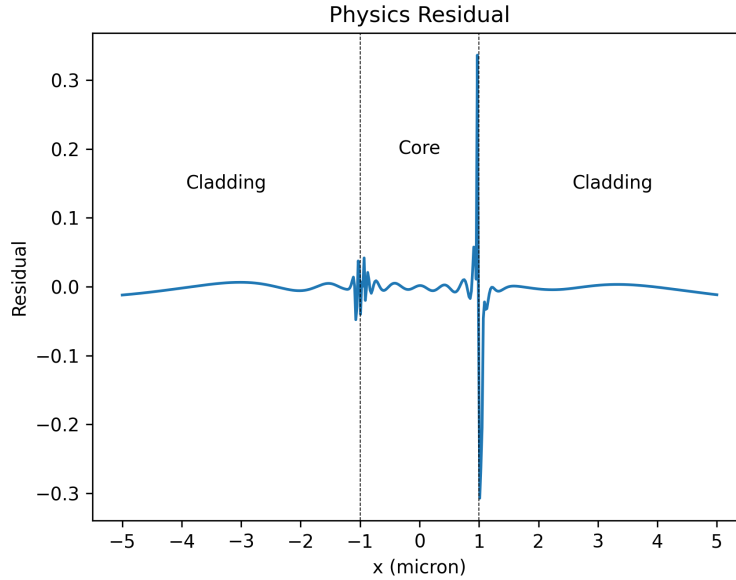


Figure 3: Physics residual across the computational domain.

The residual remains close to zero over most of the domain, with localized peaks near the core–cladding interfaces ($x = \pm 1 \mu\text{m}$). These peaks arise due to the discontinuity in

the refractive index profile, which introduces abrupt changes in the second derivative of the field. Since neural networks represent smooth functions, the interface is approximated over a narrow transition region. The localization and reduction of these peaks during training indicate improved interface resolution.

6 Validation Strategy

The performance of the proposed model is evaluated by comparison with established numerical and analytical methods. Specifically, the results obtained from the Physics-Informed Neural Network are validated against the **analytical dispersion relation of slab waveguides[2]** as well as a **finite-difference-based eigenmode solver**. These reference methods provide reliable estimates of the propagation constant and mode profile, enabling quantitative assessment of the accuracy of the learned solution.

7 Challenges

Several challenges were encountered during the implementation of the proposed approach. The **training process is sensitive to the estimation of the eigenvalue**, as the propagation constant must be learned simultaneously with the field distribution. In addition, **the optimization problem admits trivial solutions, such as the zero-field state**, which must be avoided through appropriate normalization constraints. Another practical challenge involves **ensuring training stability and achieving reasonable convergence speed**, as Physics-Informed Neural Networks often require careful tuning of hyperparameters and loss weighting to obtain accurate and stable solutions.

8 Future Work

Several extensions are planned to further enhance the scope and validation of the proposed approach. The effective index obtained from the Physics-Informed Neural Network will be quantitatively validated against analytical solutions for slab waveguides. The framework will also be extended to **enable the extraction of higher-order modes** by incorporating appropriate orthogonality or initialization strategies. In addition, **the methodology will be generalized to model two-dimensional rectangular waveguides**, allowing analysis of more practical integrated photonic structures. **Systematic parameter sweeps over core width and operating wavelength** are planned to study modal behavior under varying design conditions. Finally, the **computational efficiency of the PINN-based approach will be compared with conventional numerical eigenmode solvers** to assess its practical advantages and limitations.

9 Conclusion

A Physics-Informed Neural Network framework has been developed to solve the eigenmode problem of a slab waveguide. The model successfully predicts confined field distributions and physically consistent effective refractive indices. These preliminary results demonstrate the feasibility of using PINNs for electromagnetic eigenvalue problems in nanophotonics. Further validation and extension to higher-dimensional structures will be pursued.

References

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